Assignment_4 T-primes + 贪心

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1. 题目

`34B` : Sale

greedy, sorting, 900, https://codeforces.com/problemset/problem/34/B

```
n, m = map(int, input().split())
televisions = list(map(int, input().split()))
televisions.sort()

ans = 0

for i in range(m):
    if televisions[i] > 0:
        break
    ans += televisions[i]
print(- ans)
```

代码运行截图 （至少包含有"Accepted"）

By pydog, contest: Codeforces Beta Round 34 (Div. 2), problem: (B) Sale, **Accepted**, #, Copy

```
n, m = map(int, input().split())
televisions = list(map(int, input().split()))
televisions.sort()

ans = 0

for i in range(m):
    if televisions[i] > 0:
        break
    ans += televisions[i]
print(- ans)
```

`160A` : Twins

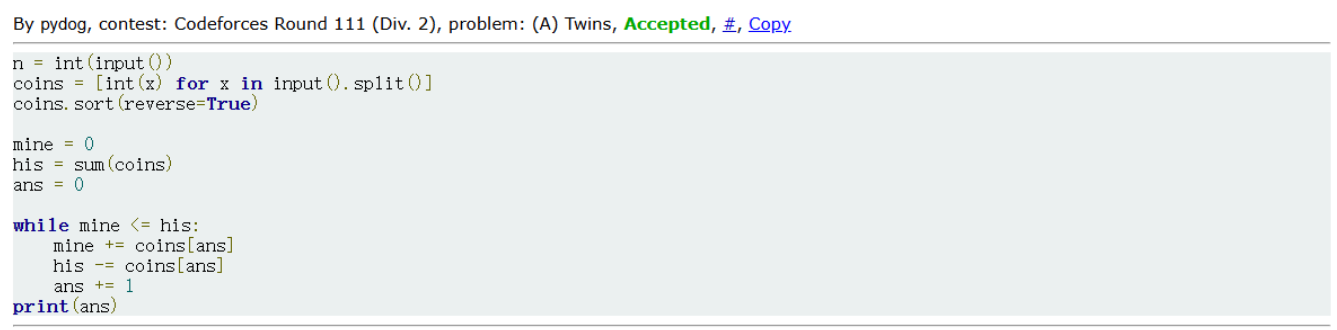
greedy, sorting, 900, https://codeforces.com/problemset/problem/160/A

```
n = int(input())
coins = [int(x) for x in input().split()]
coins.sort(reverse=True)

mine = 0
his = sum(coins)
ans = 0
```

```
while mine <= his:
    mine += coins[ans]
    his -= coins[ans]
    ans += 1
print(ans)
```

代码运行截图（至少包含有"Accepted"）



By pydog, contest: Codeforces Round 111 (Div. 2), problem: (A) Twins, **Accepted**, #, Copy

```
n = int(input())
coins = [int(x) for x in input().split()]
coins.sort(reverse=True)

mine = 0
his = sum(coins)
ans = 0

while mine <= his:
    mine += coins[ans]
    his -= coins[ans]
    ans += 1
print(ans)
```

1879B : Chips on the Board

constructive algorithms, greedy, 900, https://codeforces.com/problemset/problem/1879/B

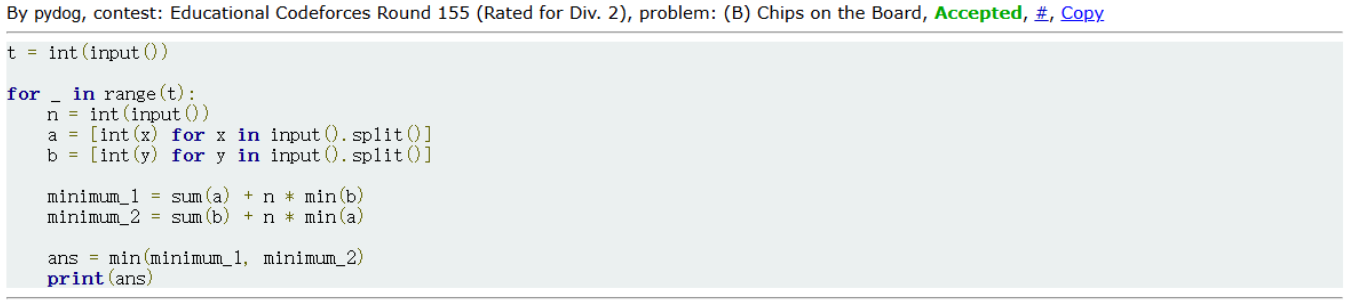
```
t = int(input())

for _ in range(t):
    n = int(input())
    a = [int(x) for x in input().split()]
    b = [int(y) for y in input().split()]

    minimum_1 = sum(a) + n * min(b)
    minimum_2 = sum(b) + n * min(a)

    ans = min(minimum_1, minimum_2)
    print(ans)
```

代码运行截图（至少包含有"Accepted"）



By pydog, contest: Educational Codeforces Round 155 (Rated for Div. 2), problem: (B) Chips on the Board, **Accepted**, #, Copy

```
t = int(input())

for _ in range(t):
    n = int(input())
    a = [int(x) for x in input().split()]
    b = [int(y) for y in input().split()]

    minimum_1 = sum(a) + n * min(b)
    minimum_2 = sum(b) + n * min(a)

    ans = min(minimum_1, minimum_2)
    print(ans)
```

M01017：装箱问题

greedy, http://cs101.openjudge.cn/pctbook/M01017/

```
while True:
    packages = list(map(int, input().split()))
    if packages == [0, 0, 0, 0, 0, 0]:
        break

    c1, c2, c3, c4, c5, c6 = packages
    ans = c6 + c5 + c4 + (c3 + 4 - 1) // 4

    u = [0, 5, 3, 1]
    available_2 = c4 * 5 + u[c3 % 4]
    if c2 > available_2:
        ans += ((c2 - available_2) + 9 - 1) // 9
    used_area = 36 * c6 + 25 * c5 + 16 * c4 + 9 * c3 + 4 * c2
    available_1 = 36 * ans - used_area
    if c1 > available_1:
        ans += ((c1 - available_1) + 36 - 1) // 36

    print(ans)
```

代码运行截图 （至少包含有"Accepted"）



`M01008` : Maya Calendar

implementation, http://cs101.openjudge.cn/practice/01008/

```
n = int(input())
print(n)

for _ in range(n):
    input_list = input().replace('.', ' ').split()
    day1, month1, year1 = int(input_list[0]), input_list[1],
int(input_list[2])

    rule_1 = {
        'pop': 1,
        'no': 2,
        'zip': 3,
        'zotz': 4,
        'tzec': 5,
        'xul': 6,
        'yoxkin': 7,
        'mol': 8,
        'chen': 9,
        'yax': 10,
        'zac': 11,
        'ceh': 12,
        'mac': 13,
        'kankin': 14,
        'muan': 15,
        'pax': 16,
        'koyab': 17,
        'cumhu': 18,
        'uayet': 19,
    }
    rule_2 = {
        1: 'imix',
        2: 'ik',
        3: 'akbal',
        4: 'kan',
        5: 'chicchan',
        6: 'cimi',
        7: 'manik',
        8: 'lamat',
        9: 'muluk',
        10: 'ok',
        11: 'chuen',
        12: 'eb',
```

```
        13: 'ben',
        14: 'ix',
        15: 'mem',
        16: 'cib',
        17: 'caban',
        18: 'eznab',
        19: 'canac',
        20: 'ahau',
    }

    absolute_date = 365 * year1 + 20 * (rule_1[month1] - 1) + day1
    year2 = absolute_date // 260
    remainder = absolute_date % 260
    a2 = remainder % 13 + 1 if remainder != 0 else 1
    b2 = rule_2[remainder % 20 + 1 if remainder != 0 else 1]

    print(a2, b2, year2)
```

代码运行截图 （至少包含有"Accepted"）



230B：T-primes（选做）

binary search, implementation, math, number theory, 1300,
http://codeforces.com/problemset/problem/230/B

```python
import math


def sieve(num: int) -> list[bool]:
    """返回布尔列表，prime_flags[i] 表示 i 是否为质数"""
    # 埃拉托斯特尼筛法
    if num < 2:
        return [False] * (num + 1)

    prime_flags = [True] * (num + 1)
    prime_flags[0] = prime_flags[1] = False

    for i in range(2, math.isqrt(num) + 1):
        if prime_flags[i]:
            for j in range(i * i, num + 1, i):
                prime_flags[j] = False

    return prime_flags


LIMIT = 10 ** 6
is_prime = sieve(LIMIT)


def is_t_prime(num):
    square = math.isqrt(num)
    if square * square == num:
        if is_prime[square]:
            return "YES"
        else:
            return "NO"
    else:
        return "NO"


n = int(input())
input_list = list(map(int, input().split()))

if __name__ == "__main__":
    for index in range(n):
        print(is_t_prime(input_list[index]))
```

也可以退而求其次

```python
import math


def linear_sieve(num: int) -> list[bool]:
    """返回布尔列表，prime_flags[i] 表示 i 是否为质数"""
    # 线性筛法/欧拉筛法
    if num < 2:
        return [False] * (num + 1)

    prime_flags = [True] * (num + 1)
    prime_flags[0] = prime_flags[1] = False

    primes = []

    for i in range(2, num + 1):
        if prime_flags[i]:
            primes.append(i)

        for p in primes:
            if i * p > num:
                break
            prime_flags[i * p] = False

            if i % p == 0:
                break

    return prime_flags


LIMIT = 10 ** 6
is_prime = linear_sieve(LIMIT)


def is_t_prime(num):
    square = math.isqrt(num)
    if square * square == num:
        if is_prime[square]:
            return "YES"
        else:
            return "NO"
    else:
        return "NO"


n = int(input())
input_list = list(map(int, input().split()))
```

```
if __name__ == "__main__":
    for index in range(n):
        print(is_t_prime(input_list[index]))
```

代码运行截图 （至少包含有"Accepted"）



By pydog, contest: Codeforces Round 142 (Div. 2), problem: (B) T-primes, **Accepted**, #, Copy

```
import math


def sieve(num: int) -> list[bool]:
    """返回布尔列表，prime_flags[i] 表示 i 是否为质数"""
    # 埃拉托斯特尼筛法
    if num < 2:
        return [False] * (num + 1)

    prime_flags = [True] * (num + 1)
    prime_flags[0] = prime_flags[1] = False

    for i in range(2, math.isqrt(num) + 1):
        if prime_flags[i]:
            for j in range(i * i, num + 1, i):
                prime_flags[j] = False

    return prime_flags


LIMIT = 10 ** 6
is_prime = sieve(LIMIT)

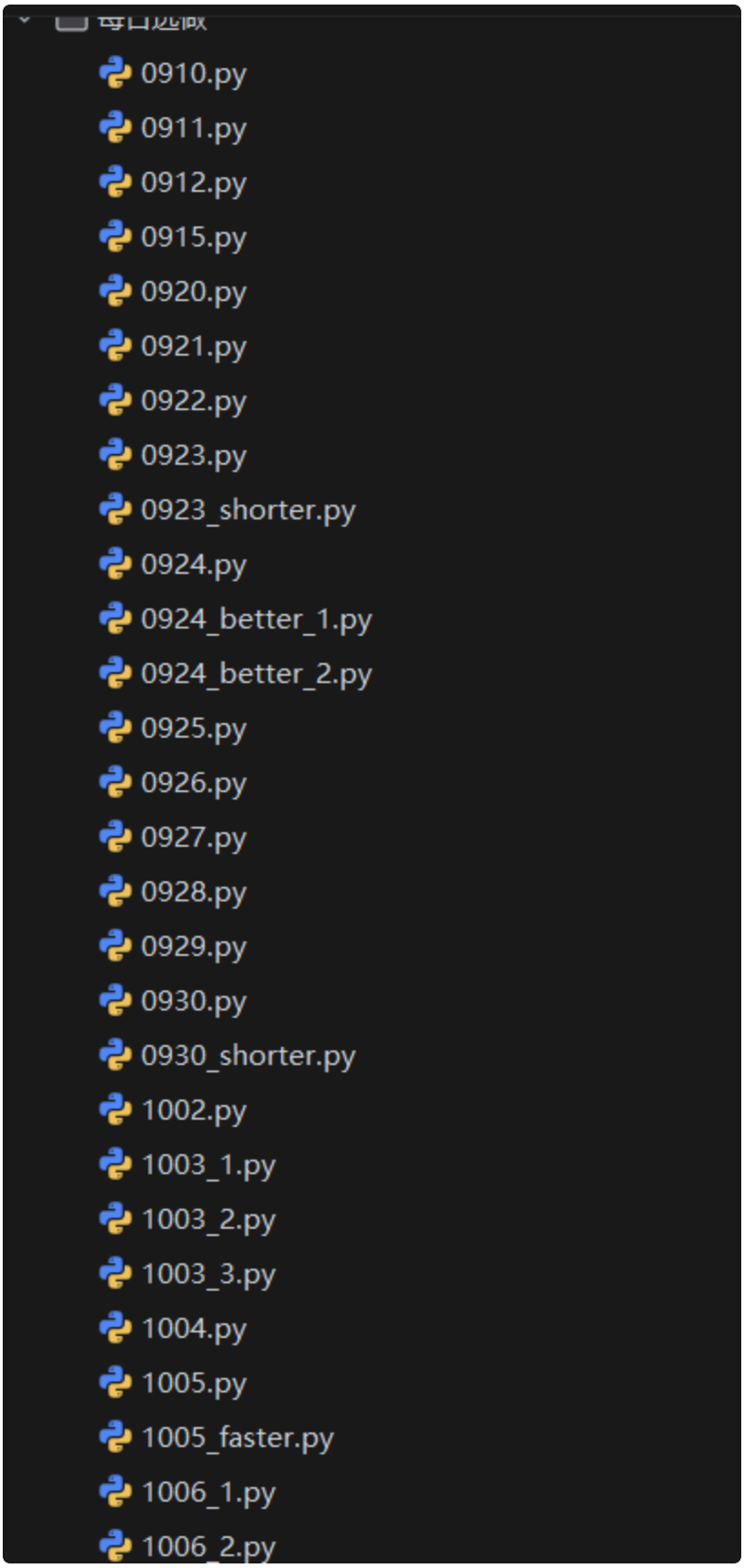

def is_t_prime(num):
    square = math.isqrt(num)
    if square * square == num:
        if is_prime[square]:
            return "YES"
        else:
            return "NO"
    else:
        return "NO"


n = int(input())
input_list = list(map(int, input().split()))

if __name__ == "__main__":
```

2. 学习总结和收获

这是我截止2026.10.6时的每日一题进度（图示题目已全部AC）



我的做题进度也已经完全同步到了Github仓库pangyouwei/2026fall-cs101上了，老师和老师的agent也可以去那里查看

感想：+：感受得到自己的小进步，有小欢喜，现在语法也牢固些了，但算法还真的没太开窍。另外自己写完能AC的代码后去看过来人写的题解的时候又会有额外的收获，比方说学到了“正则表达式”“双指针”之类的办法，虽然耗费了很多时间（包括疯狂询问DeepSeek）而且目前还不能熟练运用，但了解一下总还是好的。
-：作业题6花了几天也没想出来！看懂答案之后第一反应甚至不是喜悦，而是深深的懊恼呀，真的很痛苦，再加上我在思考的时候也想到了把$\sqrt{n}$作为一个重要分水岭去判断是否是质数，总之就是超级不爽！

现在就是一种既有信心又超级烦躁的状态，祝自己2026.10.8的小月考顺利吧！